Optimal Stratification of Survey Experiments

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Introduction

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These designs increase the **precision** of treatment effect estimation, making the most efficient use of limited experimental resources.

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1. **Sampling** of experimental participants.
2. **Assignment** of treatments to the sampled units.

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2. Assignment: how should treatments be assigned to these villages?

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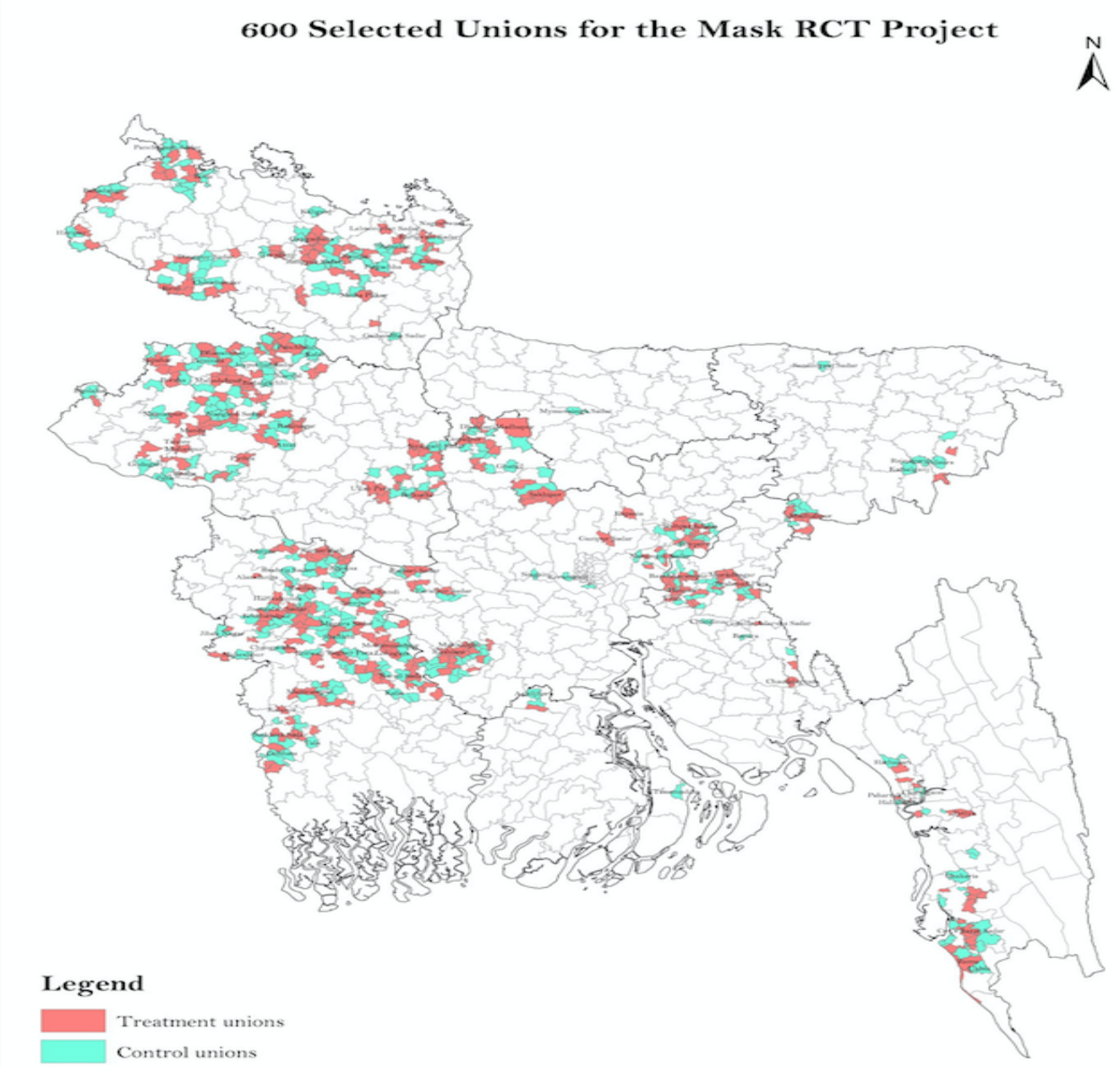

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2. Matched **k-tuples**, generalizing matched pairs to assignment propensity $p(x) \neq 1/2$ and sampling propensity $q(x)$.
3. Allows representative sampling, e.g. out of 500 villages, which 100 should you select?

Outline

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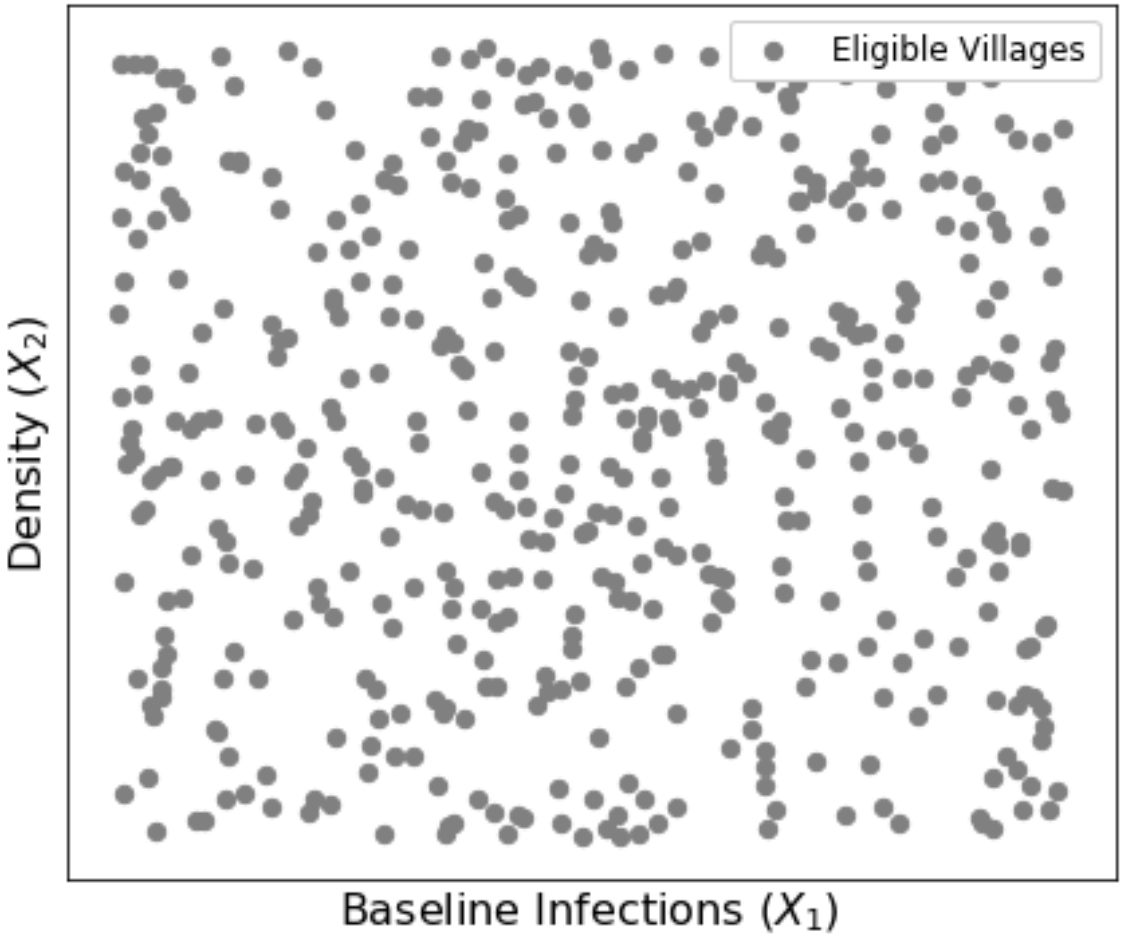
Group Homogeneity. For $\psi \in \mathbb{R}^d$ groups must satisfy

$$\frac{1}{N} \sum_g \sum_{i,j \in g} |\psi_i - \psi_j|_2^2 = o_p(1)$$

Local Randomization

$$T_{1:N} \sim \text{Loc}(\psi, 1/5) \qquad D_{1:N} \sim \text{Loc}(\psi, 2/3)$$



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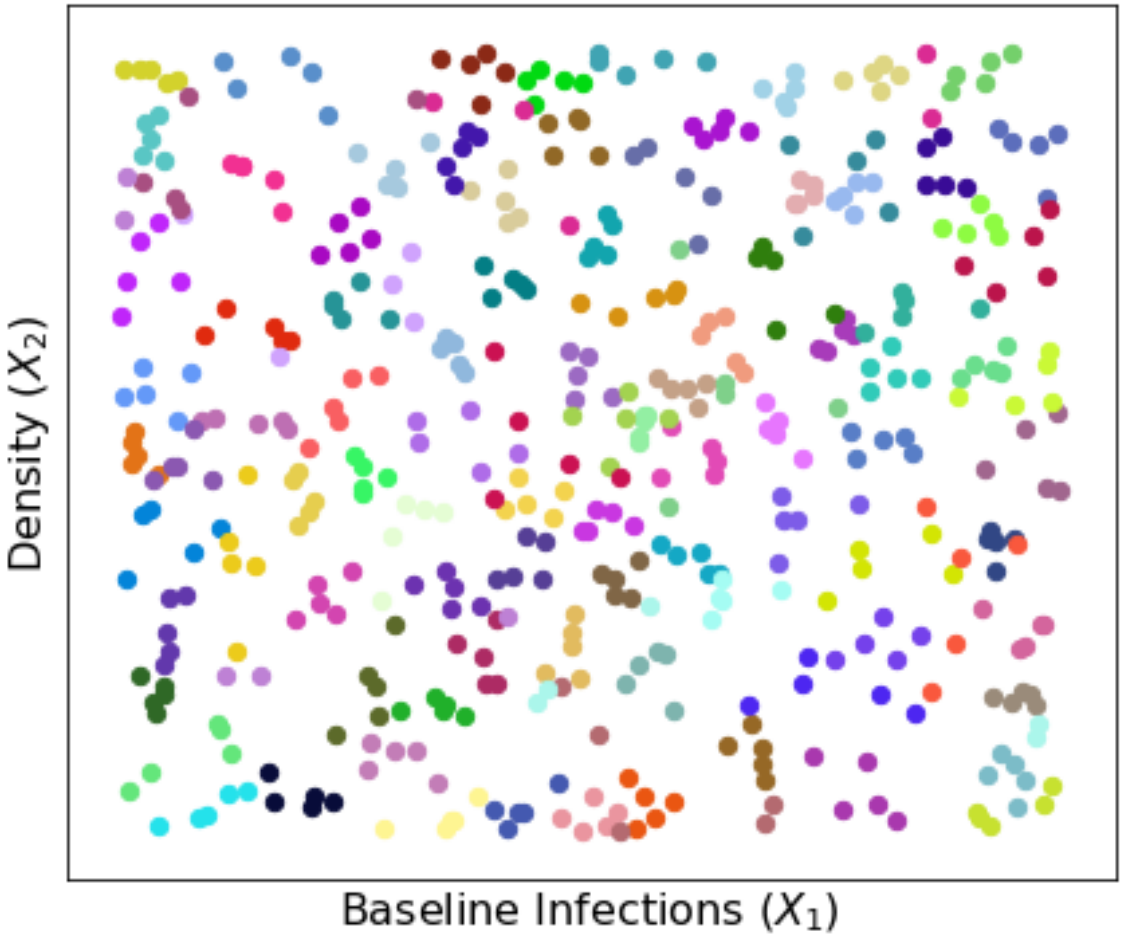

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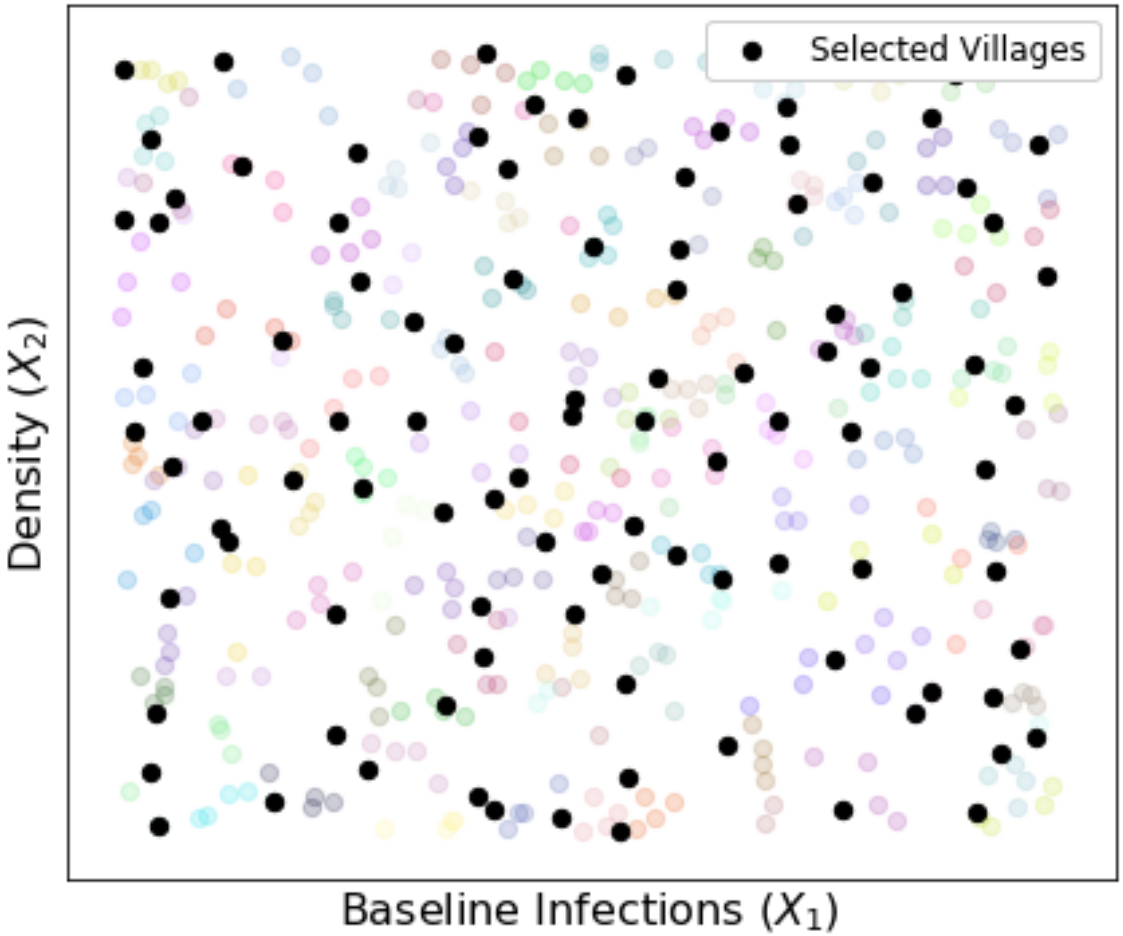

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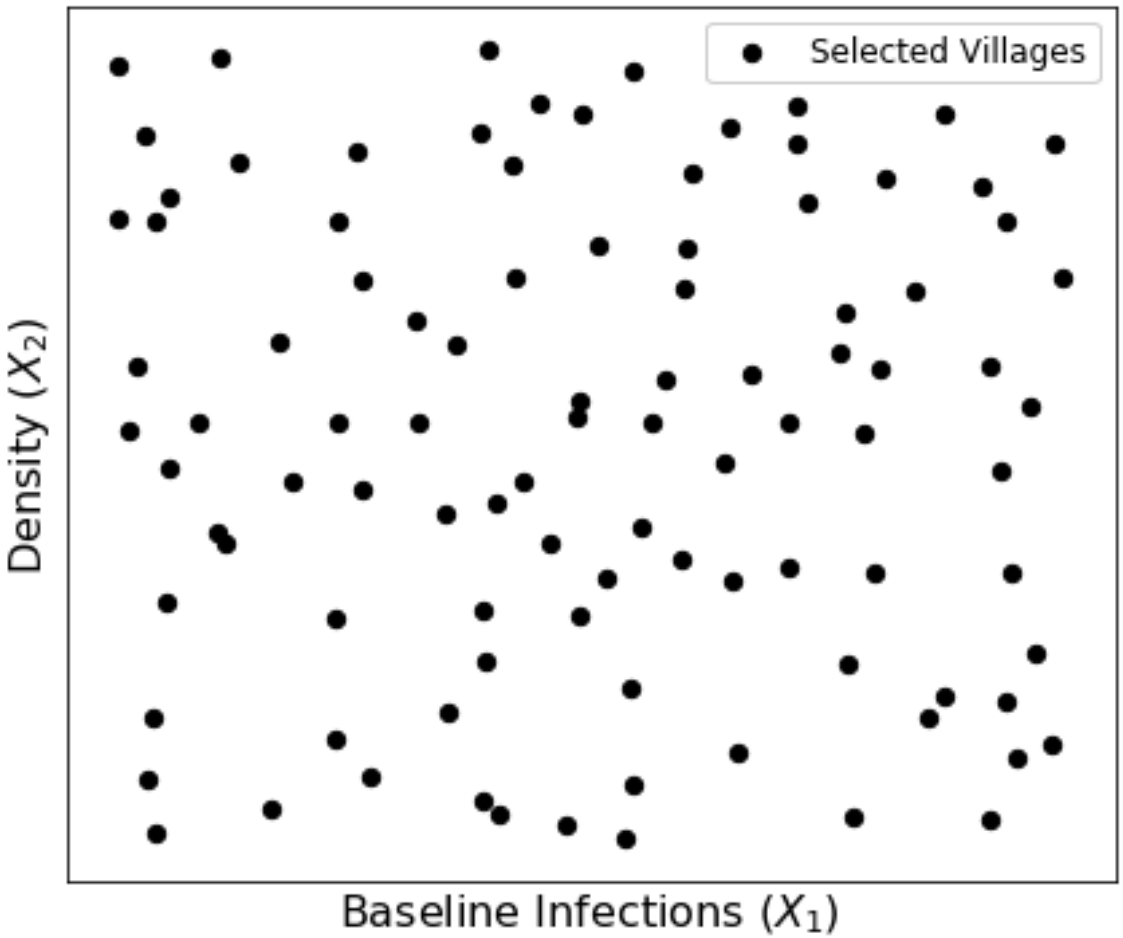

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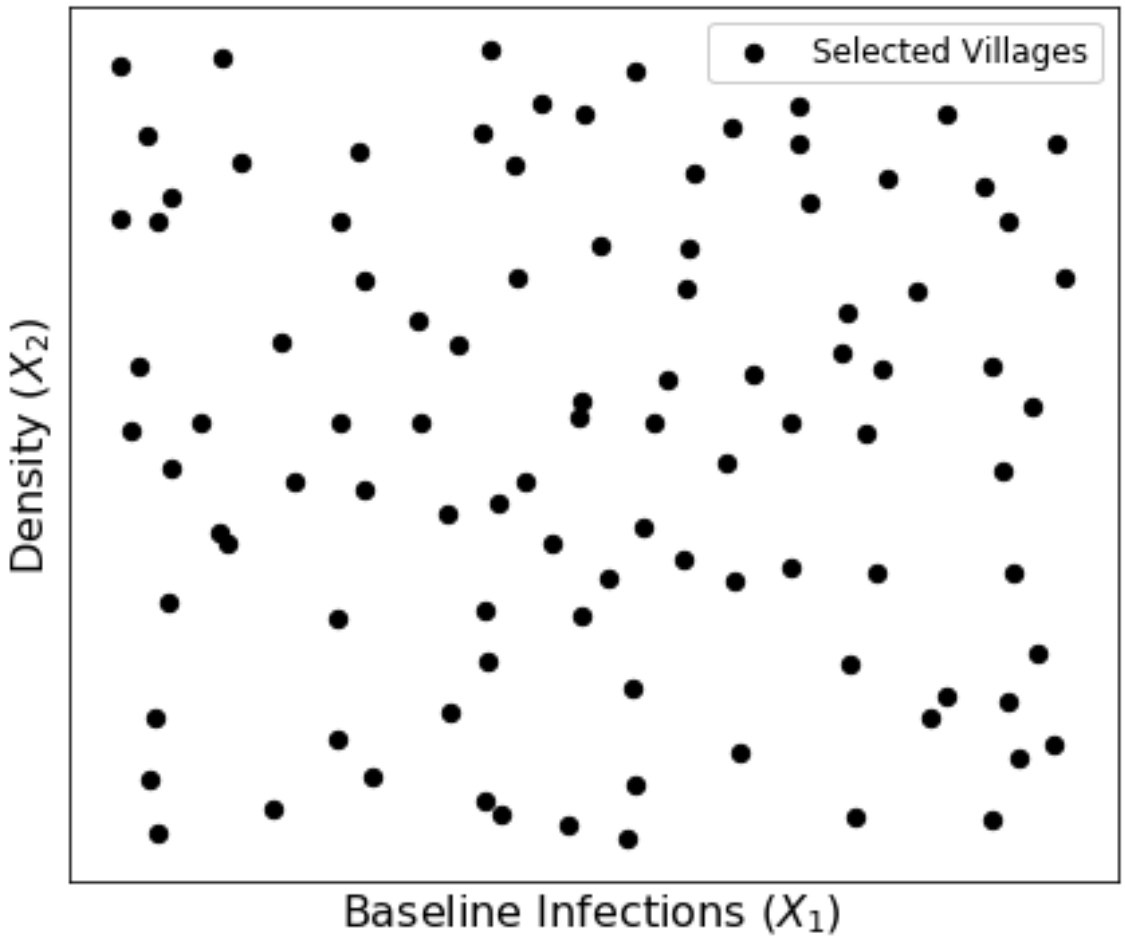

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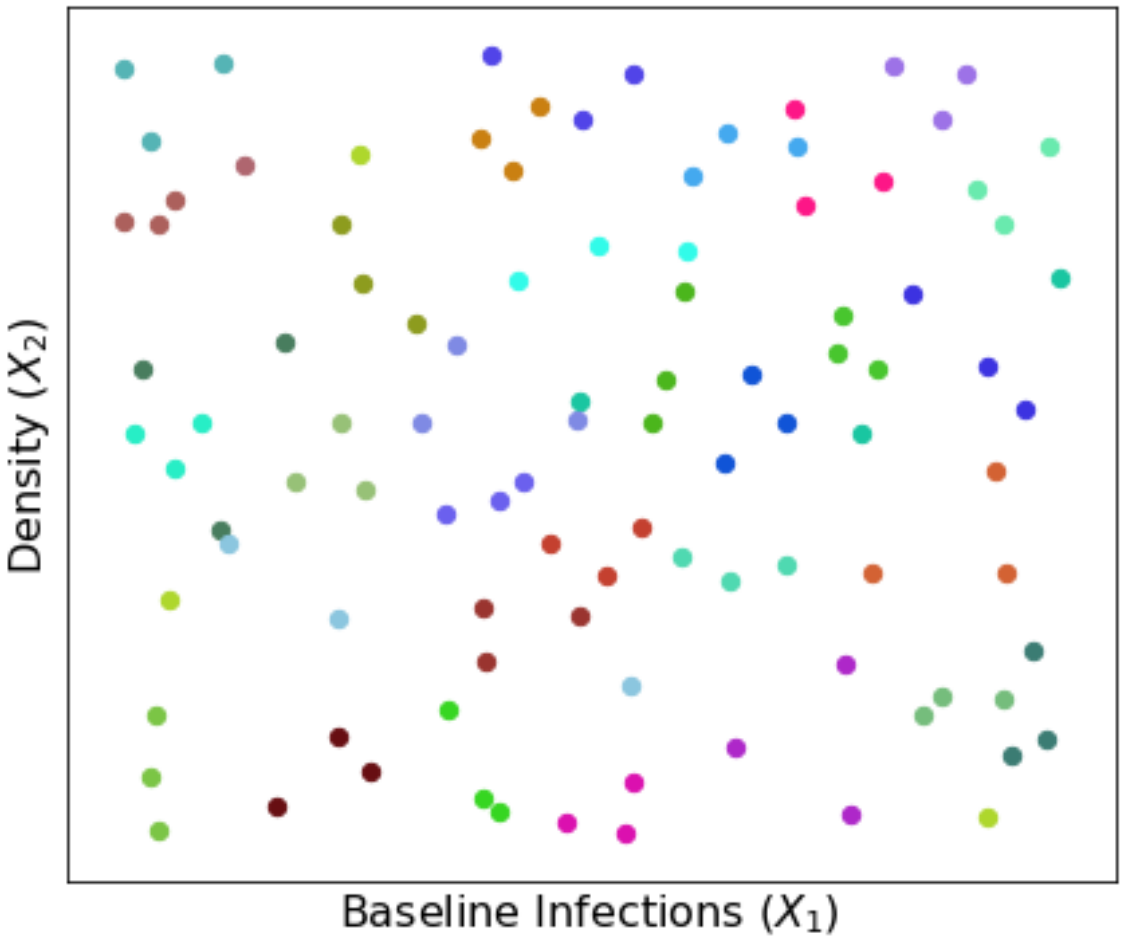

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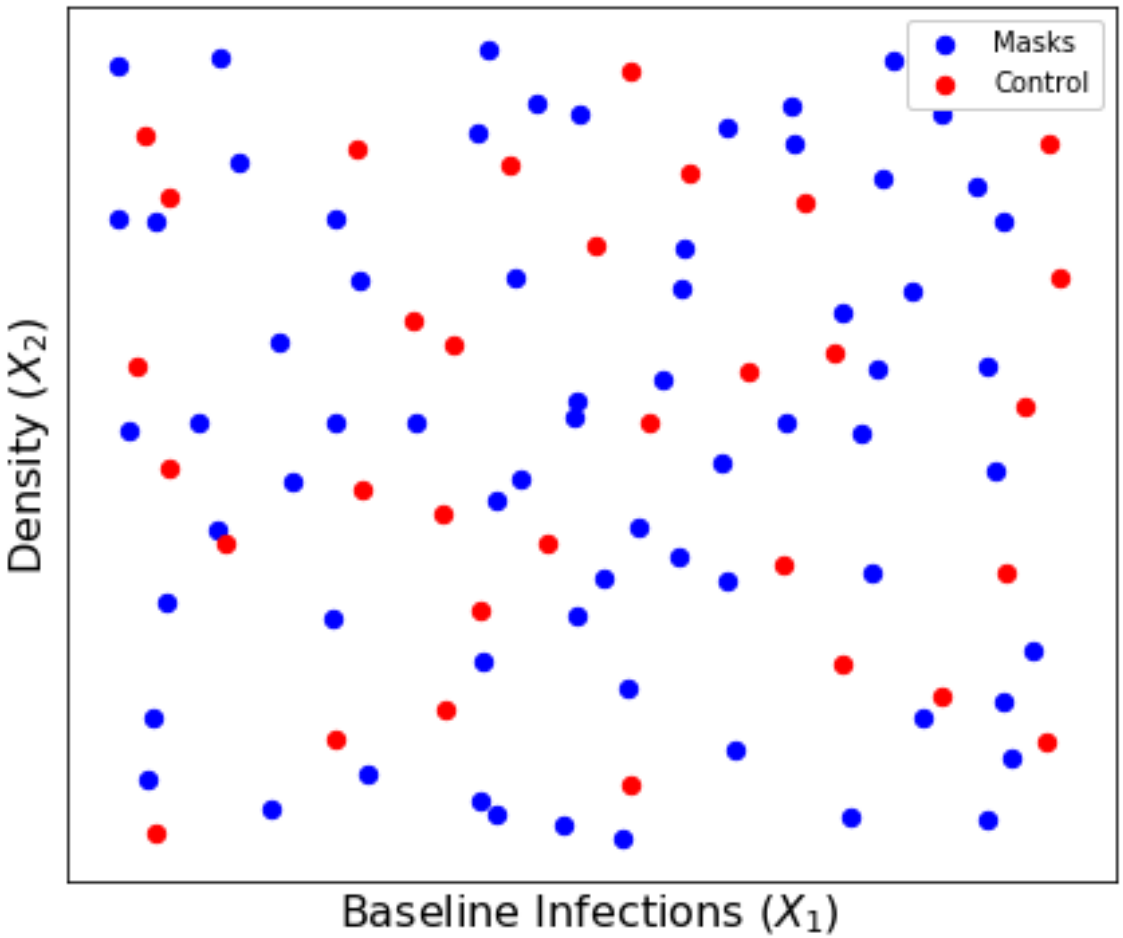

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Example. $N = 1000$ villages, can afford $n = 250$.

- Let $T_{1:N} \sim \text{Loc}(\psi, 1/4)$.
- Matched 4-tuples sampling.

ATE Estimation

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$$b(\psi; p) = \left(\frac{1-p}{p}\right)^{1/2} E[Y(1)|\psi] + \left(\frac{p}{1-p}\right)^{1/2} E[Y(0)|\psi]$$

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Variance Decomposition. Suppose $T_{1:N} \sim \mathrm{CR}(q)$ and $D_{1:N} \sim \mathrm{CR}(p)$. Then $\sqrt{n}(\widehat{\theta} - \mathrm{ATE}) \Rightarrow \mathcal{N}(0, V)$

$$V = \underbrace{\mathrm{Var}(c(\psi))}_{\text{Heterogeneity}} + \underbrace{\mathrm{Var}(b(\psi))}_{\text{Assignment}} + E\left[\frac{\sigma_1^2(\psi)}{p} + \frac{\sigma_0^2(\psi)}{1-p}\right]$$

Residual variance $\sigma_d^2(\psi) = \mathrm{Var}(Y(d)|\psi)$.

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Theorem. Suppose $T_{1:N} \sim \mathrm{Loc}(\psi, q)$ and $D_{1:N} \sim \mathrm{Loc}(\psi, p)$. Then $\sqrt{n}(\widehat{\theta} - \mathrm{ATE}) \Rightarrow \mathcal{N}(0, V)$

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If $q = 1$, this is Hahn (1998) variance bound for ATE. Nonparametric regression adjustment "by design."

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2. (2) In each propensity stratum $\{i : q(\psi_i) = a_j/k_j\}$, draw $T_{1:N} \sim \mathrm{Loc}(\psi, a_j/k_j)$.

ATE Estimation with Varying Propensities

Define the double IPW estimator

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Same variance as "double" AIPW estimator that adjusts nonparametrically for imbalances in both sampling and assignment variables.

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Leads to optimal sampling problem

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Theorem. Suppose $T_{1:n} \sim \text{Loc}(\psi, \widehat{q}_n(\psi))$ and $D_{1:n} \sim \text{Loc}(\psi, \widehat{p}_n(\psi))$. Then $\sqrt{n}(\widehat{\theta} - \text{ATE}) \Rightarrow \mathcal{N}(0, \bar{q}V^*)$.

$$\text{Var}(c(\psi)) + \min_{\substack{0 \le q,p \le 1 \\ E[C(\psi)q(\psi)] = \bar{B}}} E\left[\frac{1}{q(\psi)}\left(\frac{\sigma_1^2(\psi)}{p(\psi)} + \frac{\sigma_0^2(\psi)}{(1 - p(\psi))}\right)\right]$$

Global Optimality

Define Max-Cut problem with edge weights $w_{ij} = b(X_i)b(X_j)$

$$\max_{E_0, E_1} \sum_{i,j} b(X_i)b(X_j)\mathbb{1}(i \in E_1, j \in E_0) \quad \text{s.t.} \quad E_0 \sqcup E_1 = \{1, \ldots, n\}$$

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Theorem. The design P^* has

$$\mathrm{MSE}_{P^*}(\widehat{\theta} | X_{1:n}) \leq \mathrm{MSE}_P(\widehat{\theta} | X_{1:n})$$

for all P with $P(D_i = 1) = 1/2$ and $D_{1:n} \perp\!\!\!\perp W_{1:n} | X_{1:n}$.

Inference

Suppose $T_{1:N} \sim \text{Loc}(\psi, q(\psi))$ and $D_{1:N} \sim \text{Loc}(\psi, p(\psi))$. By earlier theorem, $\sqrt{n}(\widehat{\theta} - \text{ATE}) \Rightarrow \mathcal{N}(0, V)$.

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Theorem. (Inference) Define variance estimator

$$\widehat{V} = \mathrm{Var}_n \left(\frac{T_i(D_i - p(\psi_i))Y_i}{q(\psi_i)(p - p^2)(\psi_i)} \right) - \widehat{v}_1 - \widehat{v}_0 - 2\widehat{v}_{10}.$$

Then $\widehat{V} \xrightarrow{p} V$.

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Outline

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Draw $(\widehat{Y}_i(0), \widehat{Y}_i(1), \psi_i)_{i=1}^n$ with replacement, make design, reveal outcomes $\widehat{Y}_i = T_i(D_i \widehat{Y}_i(1) + (1 - D_i)\widehat{Y}_i(0))$, estimate and form confidence interval.

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Costs. Define costs

$$C(\psi) = \mathbb{1}(|\psi|_2 \leq \kappa) + 5\mathbb{1}(|\psi|_2 > \kappa)$$

with $\kappa = \text{Median}_{i=1}^{n} |\psi_i|_2$ and $\bar{B} = 1.5$. Results in sampling proportion $\bar{q} = E[q^*(\psi)] \approx 0.7$.

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5. **Pilot S/L**. $T_{1:n} \sim \text{Loc}(\psi, \widehat{q}_n(\psi))$ and $D_{1:n} \sim \text{Loc}(\psi, \widehat{p}_n)$ with $n_{pilot} = 100,400$.

Empirical Application

	Design, Paper	A.	Ban.	Bay.	C.	De.	Do.	F.	H.	L.
SD	CR	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
	CR, Loc	0.81	0.48	0.51	0.88	0.55	0.93	0.79	0.73	0.87
	Loc	0.75	0.45	0.39	0.84	0.47	0.88	0.74	0.70	0.76
	Hom.	0.72	0.46	0.44	0.79	0.58	0.86	0.85	0.68	0.90
	Pilot S	0.72	0.45	0.55	0.81	0.57	1.01	0.75	0.71	0.75
	Pilot L	0.70	0.42	0.52	0.81	0.52	0.87	0.65	0.68	0.71
%ΔCI	CR	0	0	0	0	0	0	0	0	0
	CR, Loc	-11	-49	-41	-6	-36	-4	-14	-10	-11
	Loc	-12	-48	-38	-5	-33	-4	-14	-15	-11
	Hom.	-21	-44	-47	-6	-40	-8	-8	-16	-10
	Pilot S	-19	-47	-23	-5	-23	3	-18	-15	-21
	Pilot L	-21	-52	-31	-5	-30	-9	-27	-18	-24
Cover	CR	0.95	0.95	0.95	0.94	0.96	0.95	0.94	0.94	0.96
	CR, Loc	0.96	0.96	0.99	0.96	0.98	0.96	0.96	0.98	0.96
	Loc	0.98	0.97	1.00	0.96	1.00	0.97	0.97	0.97	0.98
	Hom.	0.97	0.98	0.98	0.97	0.97	0.96	0.96	0.98	0.95
	Pilot S	0.97	0.98	0.99	0.96	0.99	0.95	0.97	0.98	0.97
	Pilot L	0.97	0.97	0.99	0.97	0.99	0.96	0.97	0.97	0.96
	n	1451	903	550	91	446	1000	1903	116	770
	$\dim(\psi)$	8	6	3	3	3	4	6	4	3

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2. Variance estimation reflects precision from stratified sampling and assignment, though conservative.
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4. Use $\widehat{q}(\psi)$ or $\widehat{p}(\psi)$ if have large pilot, previous observational data, large heteroskedasticity.

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Local Randomization. Generic data-adaptive fine stratification $D_{1:n} \sim \mathrm{Loc}(\psi, p(\psi))$ for $\psi \in \mathbb{R}^d$.

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